

# Appendix: protocol, retractions and reproduction

Global Quantum + AI Challenge 2026 — HSBC track

## 1 What was fixed in advance, and what the gate enforces

The pre-registration was frozen before any model was fitted and before any block other than the file header was read. Two of the guarantees claimed in the body are valid only if their parameters were chosen without reference to the data, so the commitment is mechanical rather than declarative.

**Learn-then-Test** controls family-wise error over a *finite grid fixed in advance*. Enlarging the grid after seeing which value looks good inflates the very error rate the procedure bounds. **Non-exchangeable split conformal** separately requires the weights to be fixed rather than fitted; fitting the decay on the calibration set voids the theorem.

`scripts/check_protocol.py` hashes the guarantee-bearing fields of `configs/default.yaml` — every  $\alpha$  grid,  $\delta$ ,  $\rho$ , the decision-grid size, the FWER method, the band traffic-budget and recall-floor grids, the split fractions and the decline budget — and exits non-zero on any undocumented change. It does *not* hash the protocol’s prose: correcting a mis-transcribed count should not require an amendment, and a gate irritating enough to be bypassed protects nothing. Fields entering no guarantee (output paths, fitting seeds, plotting choices) are deliberately outside the hash.

**Amendments.** The pre-registration carries 9 of them, and none was derived from  $\mathcal{D}_{\text{test}}$ . A1 and A2 come from block sizes and from  $\mathcal{D}_{\text{band}}$ ; A3 and A4 also used quantities measured on  $\mathcal{D}_{\text{cal}}$  — outer-threshold recall, and the band-conditional rate that fixes its own  $\alpha$  scale — which is disclosed rather than treated as free. The last five are one correction and four disclosures: a test specified for the wrong mechanism, a screen computed and never gated, work pre-registered and never done, a rule stated more strongly than the code enforces it, and a dataset never obtained.

- A1. The band-conditional certificate is sample-starved by construction; the reachable  $\alpha$  is capped by the band budget rather than by the method. Recorded before the arm ran, and Section 3 of the body reports the measurement landing exactly there.
- A2. PSD2 SCA-RTS rates withdrawn as a threshold selector. The ceilings are *value*-weighted, measured over a rolling 90 days across a whole portfolio, and govern exemption eligibility rather than a decline threshold.  $\mathcal{D}_{\text{band}}$ , the block this was measured on, has a value-weighted fraud rate of 5.554 %, 42.7 times the loosest ceiling of 0.13 %; meeting it on this file would require declining 94.74 % of traffic. Applying a portfolio ceiling to a fraud-enriched research benchmark is a category error, and it was ours.
- A3. The decline budget moved from 2 % to 5 %, and the recall floor is reported as a frontier rather than a fixed constraint.
- A4. The band-conditional  $\alpha$  scale, the recall-loss reformulation as a false-negative rate, and the exclusion of the decision grid’s upper endpoint.
- A5. *Dated after the certificate was produced.* H5’s pre-registered test was specified for the wrong mechanism — an exact Beta-Binomial tail is the law for split-conformal coverage, not for a Learn-then-Test certificate applied unchanged to a held-out block.
- A6. *Recorded on discovery.* The geometric-difference screen was pre-registered as an a-priori screen and computed but never gated on — a divergence between protocol and code. What rejected the arm was RBF distinctness, which admitted none of the 120 configurations; conditioning admitted 28.
- A7. *Disclosure of non-execution.* The protocol’s drift section lists five measurements as “measured, not assumed”. Three were never run — the conformal test martingale, the block permutation test, and band-conditional against marginal drift. No script exists for any of them, and the two tables they would have written do not exist.
- A8. *The single-evaluation rule described more than the code enforces.* The rule said every sweep and ablation runs on held-out slices of  $\mathcal{D}_{\text{band}}$  or  $\mathcal{D}_{\text{cal}}$ ; of the six scripts that read  $\mathcal{D}_{\text{test}}$ , four do so without the guard. Nothing was *selected* there — the ablation table feeds no claim, and every threshold comes from  $\mathcal{D}_{\text{band}}$  and  $\mathcal{D}_{\text{cal}}$  — so the guarantee stands, but the sentence did not. The protocol was corrected rather than the scripts, which would have changed committed tables to retire a wording defect.
- A9. *Disclosure of a dataset promised and never obtained.* ULB European Cardholder was pre-registered as secondary and never downloaded, so the external-validity experiment did not run and the counts quoted for it came from its published description rather than from measurement. Withdrawn rather than recomputed, which would assert a measurement not made.

**The single-evaluation rule is a mechanism, and its scope is selection.** Selection happens once: band edges and  $\lambda$  come from  $\mathcal{D}_{\text{band}}$  and  $\mathcal{D}_{\text{cal}}$  and are carried to  $\mathcal{D}_{\text{test}}$  unchanged, and nothing is tuned or ranked there. The loader records the authorised configuration hash in `results/tables/test_access.json` and raises on a second distinct one, so a reviewer can read the ledger rather than take our word for it. Amendment A8 lists the scripts that read the block for reporting without passing through it.

**Hypotheses.** H1–H5 were stated in advance with their tests. Only two were tested. H1, H2 and H3 all compare a quantum kernel against a control, and the a-priori screens rejected every kernel before any of them could run — so they are not open questions but questions the stopping rule closed. H4 — that the tensor network does not improve band-conditional average precision over the baseline — is the one the body reports, and it is reported as underpowered against its own pre-registered ceiling. H5 is validated on the held-out block.

## 2 What we got wrong

The decision log records 97 entries. These are the ones that changed a result, listed because a study that reports only its surviving claims gives a reviewer no way to judge how hard the others were pushed.

**A certificate that appeared to work was vacuous.** An early run certified a large set of configurations. Every selected  $\lambda$  sat exactly at the upper band edge — where band membership is  $f(x) < \tau_{\text{hi}}$  and the in-band rule is  $g(x) \geq \lambda$ , so the flagged set is empty by construction and the risk is structurally zero. It was the only grid point that could clear the Holm level, and it certified nothing. The decision grid now excludes the upper endpoint, and the body reports the smallest margin below  $\tau_{\text{hi}}$  (0.0136) so the property is visible rather than assumed.

**The same certificate rested on an invalid  $p$ -value.** The risk was formulated as  $\max(0, \text{floor} - \text{recall})/\text{floor}$ . Hoeffding–Bentkus bounds a mean of per-observation losses; that expression is a nonlinear transform of a mean, so the quantities it produced were not  $p$ -values and nothing resting on them was a certificate.

**The central result was narrowed twice.** A single seed’s maximum was first reported as a central estimate of the temporal inflation. Re-running across seeds gave a smaller figure with a real interval; re-running across five rolling calibration origins then showed the deviation is *origin-dependent*, with the earliest origin conservative. The clean headline did not survive, and the body reports the range instead.

**A power gate that could not fail.** The pre-registration requires the minimum detectable effect *before* the quantum comparison. The calculation had no caller at all, so H4 was going to be observed rather than tested. The fix estimated noise from the baseline against itself — correct reasoning, wrong quantity, because H4 compares model *families* and their measured standard error (0.0199) is 8.03-fold larger than the seed-to-seed figure. The gate passed a comparison it should have flagged. It now estimates cross-family noise from a second classical family (0.0164, giving a pre-registered minimum detectable effect of 0.0461) and binds on the larger of the two, and H4 is reported as underpowered against the effect the comparison could actually resolve.

**A claim asserted from theory and withdrawn when measured.** We stated that ignoring cluster structure makes intervals too narrow by roughly the square root of the cluster size. That is the classical result for a sample mean. Average precision is a rank statistic; on a fixture with cluster-level risk the clustered interval came out *narrower* than the row-level one. The row-level interval is invalid under this dependence, but the sign of its error is statistic-dependent, and we no longer claim otherwise.

**Four source files were absent from every pushed commit.** A `.gitignore` pattern without a leading slash matches at any depth, so `data/` excluded `src/hsbcfraud/data/` — the loader, the split, the test-fold guard and the label audit. `git add` does not warn when it skips an ignored path and `git status` does not list it, so the omission was invisible locally and total for anyone cloning. A hygiene test now compares the files on disk against `git ls-files`.

**Two silent bugs in the tensor network.** The initialisation placed the identity on both physical channels, making the contraction input-independent; logits were bit-identical across samples, so the AUC came out at exactly chance. Separately, closing the final bond to dimension one made renormalisation map every sample to  $\pm 1$ . Both produced plausible near-chance numbers rather than errors. A third latent hazard was removed while diagnosing a non-finite loss at 431 sites: a discarded log-norm was multiplied by zero, and  $\infty \times 0$  is not zero.

**Table 1:** The challenge statement’s primary evaluation metrics, on the held-out temporal block at the default decision threshold, for the tuned gradient-boosted baseline. Reported because the statement names all of them and a certificate is not a substitute for any of them. Recall is low because a forward holdout is a harder task than the cross-validation the published leaderboard figures come from, not because the baseline is untuned; the body measures that difference directly.

| ROC AUC | AUPRC  | $F_1$  | Precision | Recall | TP   | FP  | FN   | TN     |
|---------|--------|--------|-----------|--------|------|-----|------|--------|
| 0.8837  | 0.5114 | 0.4655 | 0.8201    | 0.3250 | 1281 | 281 | 2661 | 111311 |

### 3 Reproduction

```
make venv      # python3.12; torch from the cu130 index, which the default build lacks
make smoke    # S0-S8 environment assertions; any failure stops the build
make reproduce
make check     # rebuilds the PDFs, then the claim, citation and manifest gates
```

**Environment traps that cost real time.** `qiskit-aer` and `qiskit-aer-gpu-cu11` are distinct distributions installing the *same* package, so whichever pip unpacks last wins. Resolved together, the CPU build can land second and Aer then reports devices (`'CPU',`) with no error anywhere — a GPU cross-check silently becomes a second CPU run. The GPU wheel is reinstalled last and a smoke check asserts the device by reading `nvidia-smi`. Separately, chained assignment is a silent no-op under pandas copy-on-write now that `SettingWithCopyWarning` has been removed; a smoke check confirms the behaviour on the host and a syntactic gate forbids the pattern repository-wide.

**Measuring the simulator’s memory needed a witness outside the process.** An in-process sampler under-reported peak usage, because Aer holds the GIL and the sampling thread got two scheduling ticks across the whole run. The measurement now streams from an `nvidia-smi` subprocess and is cross-checked against Aer’s own result metadata.

**Every measured number in both documents is generated.** `docs/claims.yaml` binds each quoted figure to a table row; `scripts/check_claims.py` recomputes all of them from `results/tables/*.csv`; `scripts/make_tex.py` emits them as macros and renders the results tables from the same CSVs. `scripts/check_pdf.py` then asserts page count, paper size and font floor against the built PDF and scans the sources for numeric literals, so a typed number is a defect review can see.

**Licensing.** Code is Apache-2.0 with an SPDX header on every file, enforced by test. Raw data is never committed: IEEE-CIS is distributed under Kaggle competition rules rather than an open licence, and section 7.A of those rules restricts it to non-commercial research, which is one reason a Phase II sprint must move to the issuer’s own stream. `cuquantum-cu11` arrives under NVIDIA’s proprietary SLA and is therefore not installed by the default target; it has its own `make venv-gpu` and its own NOTICE entry. The conformal implementation was written from the published papers rather than adapted from an unlicensed repository, and that decision is recorded with its date.

### 4 The challenge statement’s primary metrics

Section 4.1 of the challenge statement names ROC AUC, AUPRC,  $F_1$ , precision, recall and a confusion matrix as the primary evaluation metrics. The body reports the first two, because they are the ones the arguments there turn on. All six are below, for the gradient-boosted baseline on the held-out temporal block at the default threshold, so that nothing the statement asks for is left to be inferred.

Precision is 0.8201 and recall 0.325, for an  $F_1$  of 0.4655 — which is what a single threshold on a 3.412% positive rate produces. That trade-off is the reason for the three-valued rule: the abstention band exists so that the decision does not have to be taken at one point on this curve, and the certificate binds the error the band commits.

These are single-threshold summaries and none of them is a guarantee. A confusion matrix describes one operating point on one block; it does not bound the error rate on the next block, which is what the body’s certificate does and what the temporal coverage result shows a point estimate cannot be trusted to do.